L3 ISIL

Course: Human–Computer Interaction / Java Swing

Instructor: Dr. Izountar Yousra

Project Theme: E-Commerce Interface Development

Instructions:

- This project must be done in groups of three/four students maximum.
- The aim is to design and develop a desktop e-commerce application using Java Swing, applying HCI (Human–Computer Interaction) principles.
- Deliverables: project mock-up, source code, executable (.jar).
- The final mark is out of 10 points, based on continuous and final evaluations.

Project Objective:

The goal of this project is to design and develop a desktop e-commerce interface using Java Swing, focusing on usability, ergonomics, and interactivity following Human–Computer Interaction principles.

Suggested Themes:

- Bookstore or library shop
- Electronics store
- Food delivery app
- Toy store for children

Weekly Work Plan (8 sessions):

1. **Session 1:** Project idea and mock-up design. *Evaluation: clarity, creativity, and use of HCI principles.*
2. **Session 2:** Project setup (packages, classes). *Evaluation: architecture quality and naming conventions.*

3. **Session 3:** Main window and navigation. *Evaluation: usability and navigation flow.*
4. **Session 4:** Product display interface. *Evaluation: layout balance and readability.*
5. **Session 5:** Shopping cart and total calculation. *Evaluation: functionality and user feedback.*
6. **Session 6:** Checkout and user form (validation). *Evaluation: clarity of messages and input handling.*
7. **Session 7:** Data storage (files or database). *Evaluation: integrity and model–interface connection.*
8. **Session 8:** Interface refinement and ergonomics. *Evaluation: visual harmony and creativity.*

Final Evaluation Grid (out of 10 points):

Criterion	**Poor (D)**	**Fair(C)**	**Good(B**	**Excellent(A)**
Project idea and mock-up design				
Code structure and organization				
Interface design and ergonomics				
Functionality and interactivity				
Presentation and originality				
Total (out of 10)	_____			